

Condensation polymers (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Represent the repeating unit of polyamides and polyesters from given monomer structures.
- Deduce the monomers for a condensation polymer given the polymer structure.

Take a quick look at the tag inside your clothing. What material is listed on the tag? Is it made up of one material or a combination? Many clothing threads are made of polymers. Some are natural polymers such as cotton, wool or rayon, and some are synthetic polymers like polyester, polyamide (also known as nylon) or spandex. But how are these polymers formed? What is the structure of their monomers? How are these polymers similar or different to addition polymers we looked at in [section S2.4.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/>)?

International Mindedness

As producing synthetic polymers fibres has become more inexpensive, the fashion industry has grown tremendously, particularly in the direction of fast fashion. Fast fashion is the rapid design, production and marketing of garments to quickly keep up with current trends using low quality materials. Some of the major issues of fast fashion are:

1. Using fossil fuels for the production of polymer fibres.
2. Clothing is being discarded rapidly due to poor quality or new trends emerging.
3. Ethical concerns about the treatment of garment workers.

The [UN's sustainable development goal](https://sdgs.un.org/goals)  (<https://sdgs.un.org/goals>) #12 is focused on responsible consumption and production. The fast fashion industry is certainly one to be considered here. When purchasing your next

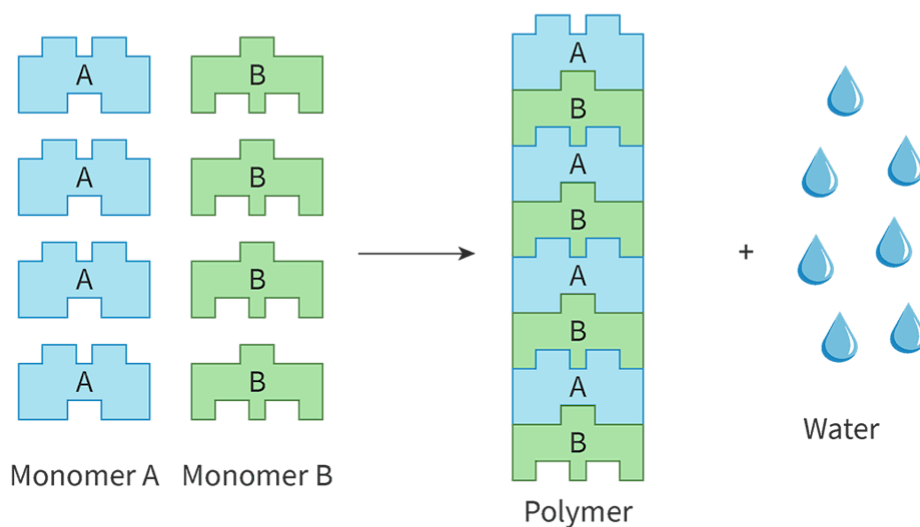
garment, consider your need, how long you intend to keep it, and its quality.

In section S2.4.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/>) you learned about addition polymers where monomers containing a double bond were able to add together without forming any byproducts, only the polymer. Most clothing fibres are instead condensation polymers. A condensation polymer is formed when two monomers react, releasing a small molecule, such as water, as a byproduct. Some condensation polymers are formed with one type of monomer and others are formed when two different monomers react and connect to each other in an alternating pattern. **Figure 1** generally depicts the reactants and products for a condensation reaction and condensation polymerisation.

Condensation reaction



Condensation polymerization with two monomers



Condensation polymerization with one monomer

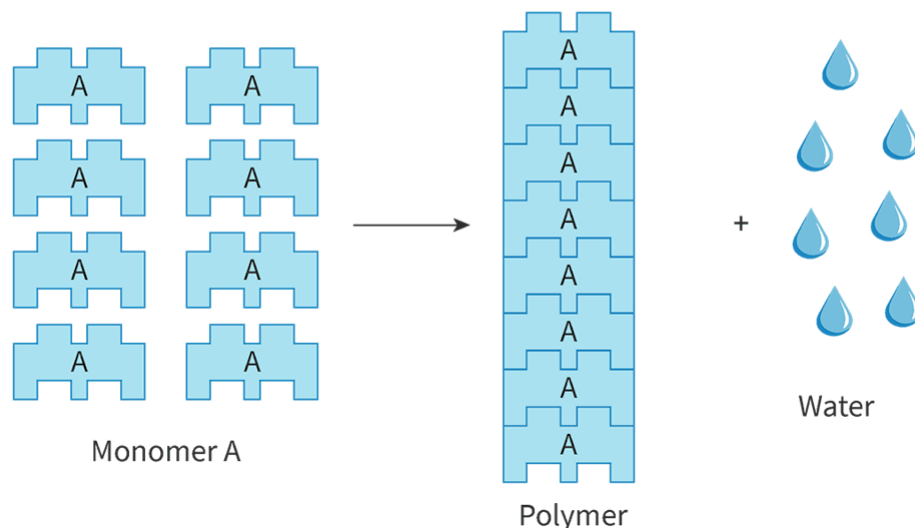


Figure 1. General condensation reaction and condensation polymerisation.

 More information for figure 1

The diagram illustrates condensation polymerization showing two different processes. On the top, it starts with two types of monomers, labeled as A and B. The monomers are depicted as puzzle pieces that fit into each other, symbolizing the reaction. Monomer A, represented in blue, and monomer B, in green, react to form a polymer chain shown in the center. Water molecules, depicted as drops, are released as byproducts during this process. The diagram includes arrows showing the process flows from monomers to polymers. Below, another representation shows polymer formation from identical monomers labeled as A forming a uniform chain without alternating types. In both processes, condensation reactions release small molecules (water) as byproducts, which is central to the concept of condensation polymerization.

[Generated by AI]

Polyesters, polyamides and biological macromolecules are all examples of condensation polymers. Condensation polymers are categorised according to the functional groups present on the repeating unit. A functional group is a group of atoms present on a molecule that give characteristic physical and chemical properties to a compound. You will learn about functional groups in greater detail in section S3.2.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/>).

Let's look at each of these categories of condensation polymers individually.

Polyester

Polyesters are most commonly known for use in fabrics and fibres. Polyesters tend to repel water. Can you think of why that would be useful in clothing? Athletic and outdoor clothing that are prone to getting wet from the environment and sweat are often made from polyester to allow for quick drying. The most common polyester found in clothing and plastics is polyethylene terephthalate (PETE).

A polyester is a polymer formed through condensation polymerisation that contains the ester functional group in its repeating unit. Let's look at the structure of the polyester PETE and identify its functional group in **Figure 2**.

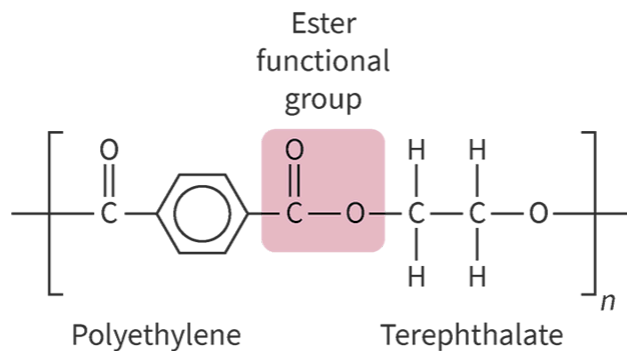


Figure 2. Polyester polymer polyethylene terephthalate.

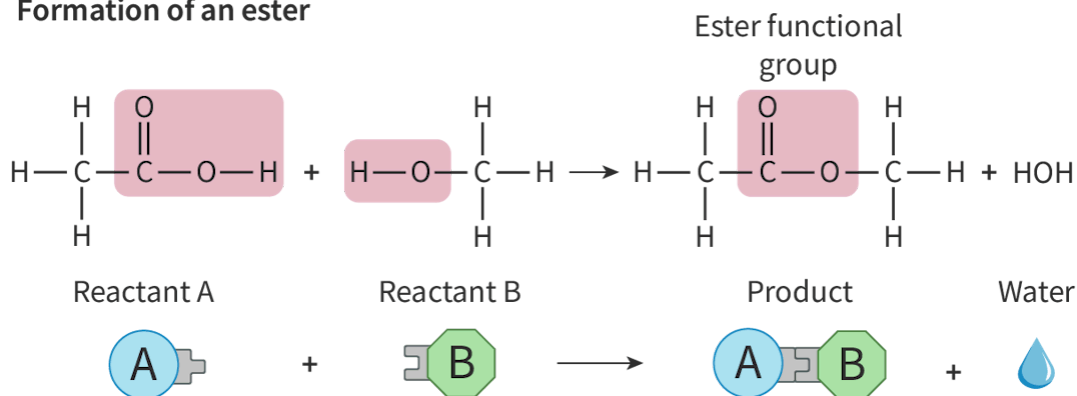
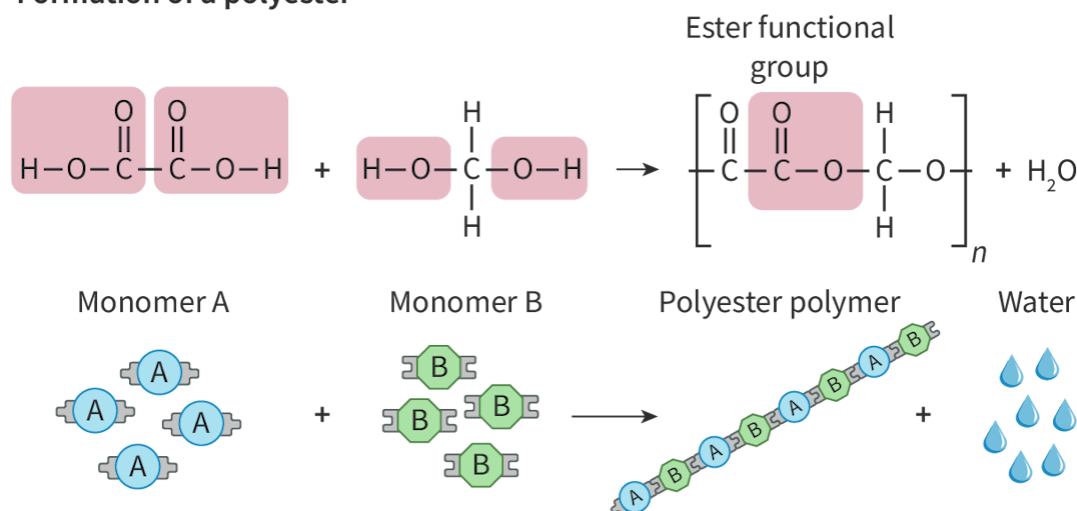
 More information for figure 2

The diagram illustrates the polyester polymer, specifically polyethylene terephthalate (PET), highlighting its ester functional group. The central part of the structure consists of a carbon atom double-bonded to an oxygen atom, and single-bonded to another oxygen atom. This second oxygen atom is further bonded to a second carbon atom, emphasizing the typical ester linkage found in polyester materials.

[Generated by AI]

Notice that the polymer in **Figure 2** has a carbon atom in the middle of the polymer chain that is bonded to two oxygen atoms, one with a double bond and one with a single bond. The oxygen with the single bond is also bonded to another carbon atom. This grouping of atoms makes up the ester functional group. You will learn more about these monomer molecules in [section S3.2.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/>). Have you noticed that this polymer is more complex in structure than the addition polymers we looked at in [section S2.4.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/>)? What do you think that means about the monomers that make up polyesters?

The ester functional group can be formed from the combination of functional groups on its reactant molecules. Polyesters are produced from the combination of monomers containing two functional groups on each monomer. **Figure 3** shows the formation of an ester and a polyester from its reactant and monomer molecules.

Formation of an ester**Formation of a polyester****Figure 3.** Formation of an ester (top) and a polyester (bottom).

More information for figure 3

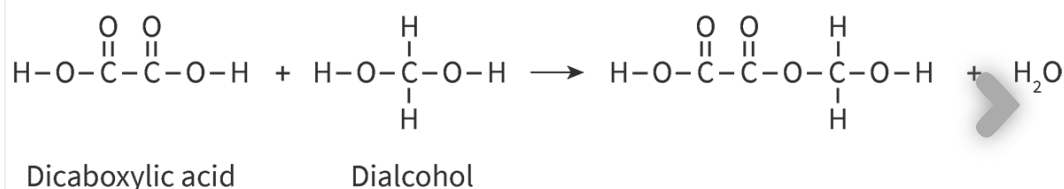
The image depicts two chemical reactions: the formation of an ester and the formation of a polyester.

1. Formation of an Ester (top part):
2. The reaction involves two molecules. On the left, "Reactant A" is shown as a molecule with a structural formula, featuring a carbon atom double-bonded to oxygen and single-bonded to an oxygen atom, forming the ester functional group, highlighted in pink. This reactant is combined with "Reactant B", which also displays a structural formula. The two reactants join, forming a "Product" shown as a combination of the two reactants, with an ester functional group at the connection point. A water molecule is also released as a byproduct, indicated by an arrow pointing to an "H₂O" symbol.
3. Formation of a Polyester (bottom part):
4. This reaction shows the combination of multiple "Monomer A" molecules and "Monomer B" molecules. Each monomer structure has an ester functional group (highlighted in pink), similar to that in the ester formation. "Monomer A" and "Monomer B" react to form a long chain, the "Polyester polymer", depicted as a repeated sequence of A-B-A-B units. Water

molecules are represented as byproducts of the reaction, illustrated as multiple water droplets.

[Generated by AI]

Notice in **Figure 3** how water is produced as a byproduct. The release of this small molecule, water, is why reactions of this type are referred to as condensation reactions. **Interactive 1** goes through more specifics about the type of reaction, including the names of some families of compounds you will learn about in more detail in section S3.2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/>). Click the arrows to view the slides.



Interactive 1. Formation of a polyester polymer.

 More information for interactive 1

A slideshow with six slides displays the Formation of a Polyester Polymer. Slide 1. The chemical reaction between a dicarboxylic acid and a dialcohol. The dicarboxylic acid is represented as a molecule with two carboxylic acid groups (–COOH) on either end.

The structure starts with a hydrogen atom bonded to an oxygen (hydroxyl group), followed by a carbon double bonded to an oxygen (carbonyl group). This carbon is connected to another carbon with the same carboxylic acid structure and its structural formula is

HOOC–COOH. The acid is labelled as Dicarboxylic acid under it. The dialcohol is depicted as a central carbon atom bonded to two hydroxyl groups (–OH) and two hydrogen atoms. The chemical formula of the alcohol is HO–CH₂–OH. In this case, the central carbon is shown with one hydroxyl on each side. The molecule is labelled as dialcohol under it. An arrow points to the products formed: an ester molecule and water (H₂O). The ester molecule formed by linking the dicarboxylic acid and the dialcohol. The left part of the molecule remains the same as the dicarboxylic acid, but one –OH group

from the dicarboxylic acid has reacted with the $-OH$ from the dialcohol, forming a new ester bond ($-\text{CO}-\text{O}-$). The structure preserves the $-OH$ of the diol on the other end. A single water molecule is shown as a byproduct of the condensation reaction.

Slide 2: The $-OH$ from the right side of dicarboxylic acid and the H on the left side of the dialcohol are highlighted. The reaction of the groups leads to the formation of water on the product side, and the water molecule is also highlighted. On-screen text reads: Water is formed from the $-OH$ of the functional group on the carboxylic acid combining with the H of the functional group on the alcohol.

Slide 3. The COOH and OH groups of the ester product are highlighted. On-screen text reads: Notice how the polymer formed still has the functional groups of the alcohol and carboxylic acid. This allows it to react further.

Slide 4. A chemical equation shows the condensation reaction of three organic molecules forming a larger molecule and producing two H_2O molecules. The reactants include three dialcohol, $\text{HO}-\text{CH}_2-\text{OH}$, on the left side, the ester product in the middle and the dicarboxylic acid $\text{HOOC}-\text{COOH}$ on the right side. The H atom in dialcohol, the $-OH$ group on the left side of ester and H on the right side of ester along with the $-OH$ group on the left side of the dicarboxylic acid are highlighted. The product formed is a polyester after the loss of two water molecules from the reactant molecules. On-screen text reads: Now two H_2O molecules have been produced.

Slide 5. A single ester molecule is shown, consisting of repeating units of carbon, hydrogen, and oxygen atoms. The structure begins and ends with a hydroxyl group ($\text{H}-\text{O}$) bonded to a carbon atom. The structure is given as, $\text{H}-\text{O}-\text{CH}_2-\text{O}-\text{CO}-\text{COO}-\text{CH}_2-\text{O}-\text{CO}-\text{CO}-\text{O}-\text{H}$. The overall chain shows multiple ester groups, indicating a polyester structure formed through condensation polymerisation. On-screen text reads: The product formed still has the functional group of an alcohol and carboxylic acid. This allows it to continue to react to form a polymer.

Slide 6. Chemical equation of polyester formation from monomers. On the left side,

n $\text{HOOC}-\text{COOH}$ under it, the name dicarboxylic acid monomer is written; the next reactant is

n $\text{HO}-\text{CH}_2-\text{OH}$ under it, the name dialcohol monomer is written. The reaction forms a repeating polymer unit in brackets, labelled as polyester polymer, with linkage n $[-\text{CO}-\text{COO}-\text{CH}_2-\text{O}-]$ and water molecule is represented as $(2n-1)$ H_2O labelled as Water by product.

The interactive helps users understand the steps involved in polyester formation.

The video shows the formation of polyesters in a bit more detail. It is worth revisiting this video once you have studied the family of compounds that make up these monomers [section S3.2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/).

Organic Condensation Polymers 1. Polyesters



Video 1. Formation of polyester polymers.

 More information for video 1

1
00:00:00,080 --> 00:00:00,920
[silence]

2
00:00:01,320 --> 00:00:04,880
instructor: Here I take a look
at condensation polymers

3
00:00:05,080 --> 00:00:07,400
or condensation polymerization.

4
00:00:07,960 --> 00:00:10,960
There are two main categories
of condensation polymer,

5
00:00:12,200 --> 00:00:14,760
polyesters and polyamides.

6
00:00:14,840 --> 00:00:17,080
So here I take a look at polyesters.

7
00:00:19,200 --> 00:00:22,280
Polyesters are

made from two monomer units.

8

00:00:22,800 --> 00:00:25,800

Here we see a so-called
dicarboxylic acid.

9

00:00:26,800 --> 00:00:27,920

We have a molecule

10

00:00:28,000 --> 00:00:32,480

that has a carboxylic acid
or carboxyl group at both ends.

11

00:00:33,320 --> 00:00:35,320

The bit that's highlighted in green

12

00:00:35,400 --> 00:00:37,600

between the carboxyl groups
can vary,

13

00:00:38,080 --> 00:00:40,320

so we can have

lots of different possibilities,

14

00:00:40,440 --> 00:00:43,080

but we'd need to have
a carboxyl group at either end

15

00:00:43,280 --> 00:00:46,320

of the dicarboxylic acid, as you'll see.

16

00:00:47,280 --> 00:00:49,560

And the second monomer
is a diol,

17

00:00:49,640 --> 00:00:55,200

a molecule with alcohol or OH
or hydroxyl groups at both end,

18

00:00:55,280 --> 00:00:56,400

so-called diol.

19

00:00:56,720 --> 00:00:59,880

And here again, the bit and blue can vary,

20

00:01:00,000 --> 00:01:02,200

but we must have an OH at both ends.

21

00:01:02,520 --> 00:01:04,240

And this is how they interact.

22

00:01:04,640 --> 00:01:06,800

The OH of the carboxyl group,
23

00:01:07,360 --> 00:01:10,440

including you notice the bond
between the C and the O
24

00:01:10,720 --> 00:01:14,560

that comes away
from the carboxyl group and the H.
25

00:01:15,040 --> 00:01:20,240

The H only comes away
from the OH of the adjacent diol.
26

00:01:20,800 --> 00:01:22,680

Together they make a molecule of water
27

00:01:23,480 --> 00:01:25,840

and then the two electrons
that represent the bond
28

00:01:25,920 --> 00:01:28,400

that had been between the O
and the H of the hydroxyl group.
29

00:01:28,760 --> 00:01:31,400

Those two electrons are used
to make a bond
30

00:01:31,480 --> 00:01:33,800

with the C of the carboxyl group
31

00:01:34,880 --> 00:01:36,560

that's known as esterification,
32

00:01:36,640 --> 00:01:41,120

the interaction of a carboxyl group
with a hydroxyl group,
33

00:01:41,640 --> 00:01:43,600

a carboxylic acid with an alcohol.
34

00:01:45,960 --> 00:01:48,720

And if we do that
at both ends of these molecules,
35

00:01:48,800 --> 00:01:53,360

in other words,
take an OH as a hydroxide ion off
36

00:01:54,280 --> 00:01:55,800

both carboxyl groups

37

00:01:56,200 --> 00:01:59,400

and take H as a hydrogen ion off both

38

00:01:59,840 --> 00:02:01,480

hydroxyl groups of the diol.

39

00:02:02,160 --> 00:02:04,200

Then this is what we should have

40

00:02:05,200 --> 00:02:08,400

and let's use those two electrons

of what had been the bond

41

00:02:08,480 --> 00:02:10,280

between the O and the H of the hydroxyl.

42

00:02:10,360 --> 00:02:11,840

To form that linkage,

43

00:02:12,800 --> 00:02:16,600

we're joining up

what had been the CO of the diol

44

00:02:17,120 --> 00:02:20,040

with the C double bond O

of what had been the carboxyl group

45

00:02:20,800 --> 00:02:22,840

and that's known as an ester link.

46

00:02:24,040 --> 00:02:27,880

I've drawn around

the ester linkage C double bond O,

47

00:02:27,960 --> 00:02:31,520

dash O dash C,

which you find in all esters.

48

00:02:32,400 --> 00:02:35,720

And what we do is

we do that on a large scale,

49

00:02:36,920 --> 00:02:40,080

taking a large number

of both types of molecule

50

00:02:40,560 --> 00:02:42,160

and alternating them like this.

51

00:02:42,320 --> 00:02:43,760

So there I've got

52

00:02:44,080 --> 00:02:45,920

the one from the dicarboxylic acid

53

00:02:46,080 --> 00:02:50,080

and I link up the one

from the diol and keep alternating.

54

00:02:50,160 --> 00:02:53,040

There's another one has come in

from the dicarboxylic acid

55

00:02:53,560 --> 00:02:54,760

and one from the diol.

56

00:02:54,840 --> 00:02:58,720

So you see that we're using both ends

of both molecules,

57

00:02:58,800 --> 00:03:02,320

which is why we need

a dicarboxylic acid and a diol.

58

00:03:05,400 --> 00:03:08,200

Remember the middle bit

there highlighted in green.

59

00:03:08,600 --> 00:03:11,920

That can vary from one

dicarboxylic acid to another.

60

00:03:12,080 --> 00:03:14,600

So it means you can form

lots of different polyesters,

61

00:03:15,000 --> 00:03:18,920

as is the case with the blue bit

highlighted for the diol.

62

00:03:20,520 --> 00:03:22,680

So varying the green,

varying the blue

63

00:03:23,120 --> 00:03:27,040

gives rise to lots

of different polyesters.

64

00:03:27,440 --> 00:03:29,840

Then in the abbreviated
form of this reaction,

65

00:03:29,960 --> 00:03:31,600

as I'm drawing out here,

66

00:03:31,800 --> 00:03:37,520

we take n dicarboxylic acid molecules
and react them with n diol molecules,

67

00:03:38,000 --> 00:03:41,920

where n 's a large whole number
or large integer, such as 1 or 2000.

68

00:03:43,040 --> 00:03:45,720

And that means that we'll end up
with n of these

69

00:03:45,800 --> 00:03:48,720

showing how they're joined
through those ester linkages.

70

00:03:50,040 --> 00:03:52,720

Now let's figure out how many
water molecules will be produced.

71

00:03:53,240 --> 00:03:54,600

Think of this analogy.

72

00:03:54,680 --> 00:03:58,280

If you get fence posts,
you have seven spaces between them.

73

00:03:58,880 --> 00:04:02,320

So in this case, hopefully you can see
that if you've got $2n$ molecules

74

00:04:02,400 --> 00:04:05,680

in total you'll have

$2n$ minus 1 water molecules made.

75

00:04:06,200 --> 00:04:09,520

But notice that I've also taken
effectively another water molecule off

76

00:04:10,240 --> 00:04:13,160

since I've taken an OH off
the left and the H off the right.

77

00:04:13,240 --> 00:04:17,280

So there is actually a total of $2n$

water molecules produced in that scenario.

78

00:04:17,760 --> 00:04:19,800

But if you draw it like this

at the bottom left,

79

00:04:19,880 --> 00:04:21,840

where you've left the OH on the left

80

00:04:21,960 --> 00:04:24,040

and left the H on the extreme right,

81

00:04:24,520 --> 00:04:27,800

then in that scenario,

you do have to indicate

82

00:04:28,200 --> 00:04:31,720

that you get $2n$ minus 1

water molecules instead.

83

00:04:32,880 --> 00:04:36,080

So that's my look

at the formation of a polyester.

84

00:04:37,000 --> 00:04:38,920

I felt I needed to show both alternatives

85

00:04:39,000 --> 00:04:42,440

because you'll see both

in various textbooks.

86

00:04:44,240 --> 00:04:47,400

One of the accompanying videos takes

a look at a specific polyester

87

00:04:47,480 --> 00:04:48,640

called terylene.

88

00:04:49,280 --> 00:04:50,760

Perhaps you'll consider looking at that.

89

00:04:50,840 --> 00:04:52,280

Thanks for listening!

Polyamide

Polyamides form fibres that are strong allowing it to be used in products such as ropes, seatbelts, camping equipment, carpets and dental floss. Some of the most common polyamides are often referred to as nylon. A polyamide is a polymer formed

through condensation polymerisation that contains the amide functional group in its repeating unit. Let's take a look at the structure of nylon-6,6, a common polyamide used in carpets, in **Figure 4** and identify its functional group.

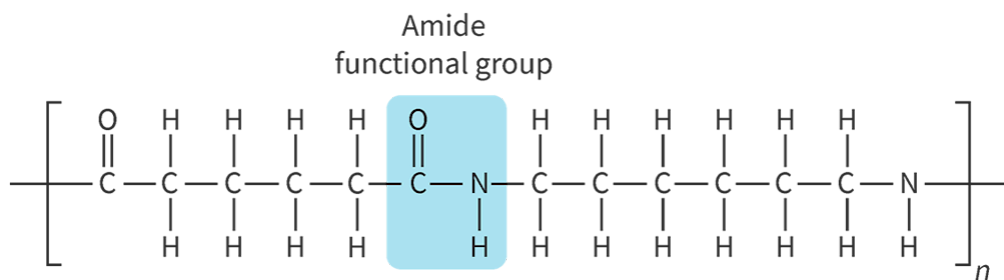


Figure 4. Polyamide polymer nylon-6,6.

 More information for figure 4

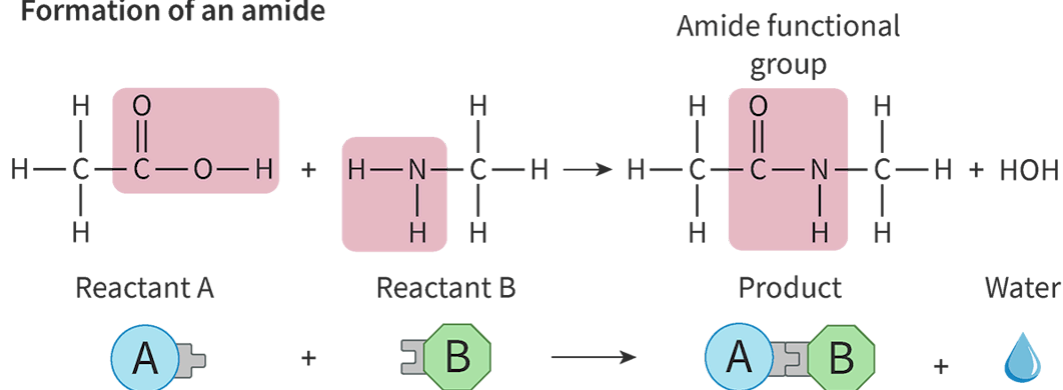
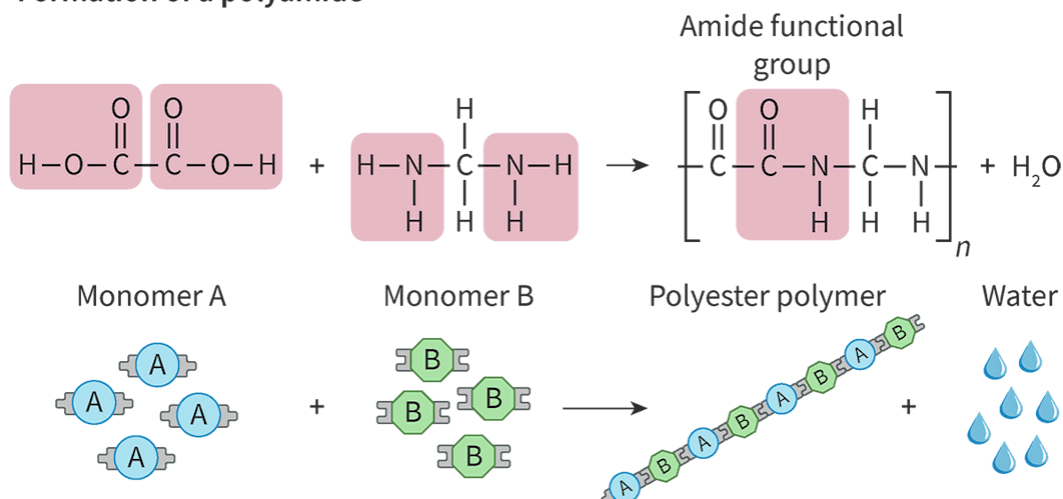
The image displays the chemical structure of the nylon-6,6 polymer. The structure is represented in a linear format showing repeating units where the amide functional group is highlighted. This functional group consists of a carbon atom doubly bonded to an oxygen atom and singly bonded to a nitrogen atom, which in turn is bonded to a hydrogen atom. The carbon atom is also part of a chain link with other carbon atoms, forming the backbone of the polymer. The repeating units emphasize the polyamide characteristic of nylon-6,6, displaying the connectivity that allows it to form long, strong fibers useful in various products such as ropes and carpets.

[Generated by AI]

Notice that the polymer in **Figure 4** has a carbon atom in the middle of the polymer chain that is doubly bonded to one oxygen atom and singly bonded to a nitrogen atom. This grouping of atoms makes up the amide functional group. Can you think of any reason why the polymer in **Figure 4** is known as nylon-6,6?

Notice how each side of the nitrogen functional group is bonded to a chain containing 6 carbon atoms? This is where the naming of different types of nylon come from.

Polyamides are produced in a very similar manner to polyesters requiring monomers that contain two functional groups. The functional groups on one of the monomers differs from those that make up polyesters. **Figure 5** shows the formation of an amide and a polyamide from its reactant and monomer molecules.

Formation of an amide**Formation of a polyamide****Figure 5.** Formation of an amide and a polyamide.

More information for figure 5

The diagram illustrates the chemical process of forming an amide and a polyamide from monomers. It is organized into three main sections, showing both molecular structures and the reactions involved.

1. Amide Formation:

- On the top left, a chemical structure is depicted showing an amide bond formation between a carboxylic acid group ($\text{O}=\text{C}-\text{OH}$) and an amine group (NH_2). The connection releases water (depicted by a water droplet icon) as a byproduct.

3. Polyamide Formation:

- On the right, two different monomers labeled as 'A' and 'B' each possess distinct functional groups. Monomer 'A' has a blue circle with an arrow indicating it connects with monomer 'B', which is shown in a green pentagon.
- The structure displayed in the mid-section depicts the repeated sequence of amide bonds forming a chain, denoting polyamide structure. Arrows indicate the continuous sequence of reactions, creating repeating units (A-B-A-B). Every connection releases a water molecule during the process.

6. Polyamide Chain Representation:

7. The bottom section further elaborates the polyamide formation by showing elongated chains of monomers. Each chain consists of alternating 'A' and 'B' units, emphasizing the polymer's repeating structure.

[Generated by AI]

The video shows the formation of polyamides in a bit more detail. It is worth revisiting this video once you have studied the family of compounds that make up these monomers in section S3.2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/>).

Organic Condensation Polymers 3. Polyamides



Video 2. Formation of polyamide polymers.

The video is an example of producing a polyamide polymer, nylon-6,10 in a laboratory setting.

Nylon Synthesis Chemistry Demo



Video 3. Production of polyamide nylon-6,10.

Theory of Knowledge

The widespread use of synthetic fibres in the clothing industry began in the 1940s as a more durable and cost-effective alternative to natural fibres. Greater durability also means greater resistance to breaking down, resulting in millions of tonnes of global textile waste each year as clothing styles go in and out of fashion. To what extent is the creator and the user of this technology responsible for the impact on the environment?

Biological macromolecules

All biological macromolecules are formed through condensation reactions. Some examples of biological macromolecules that are also polymers are:

- polypeptides (proteins)
- polysaccharides (sugars)
- polynucleotides (nucleic acids such as DNA).

The structure of these polymers and their monomers are shown in **Figure 6**.

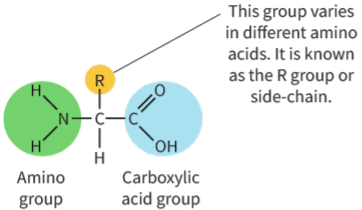
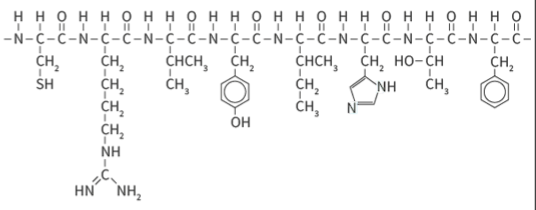
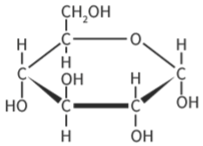
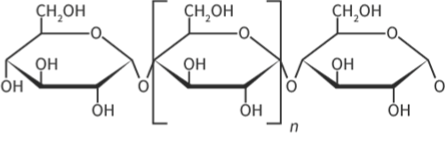
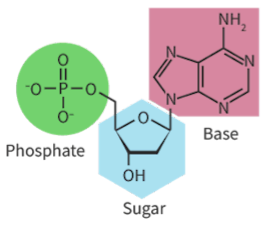
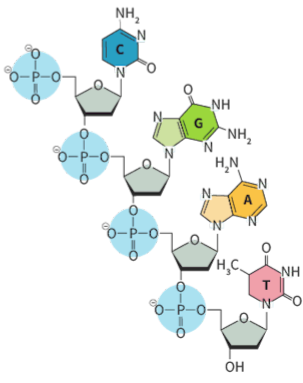
	Monomer	Polymer
Polypeptides	 Amino acid	 Polypeptide
Polysaccharides	 Saccharide	 Polysaccharide
Polynucleotides	 Nucleotide	 Polynucleotide

Figure 6. Monomer and polymer structures for some biological macromolecules.

 More information for figure 6

The image is a diagram showcasing the structures of monomers and polymers for three types of biological macromolecules: polypeptides, polysaccharides, and polynucleotides. The diagram is divided into three rows, each representing one type of macromolecule.

1. Polypeptides:
2. Monomer: The left section shows an amino acid structure featuring an amino group and carboxylic acid group with a variable R group (side chain).
3. Polymer: The right section displays a polypeptide chain with repeated units of amino acids connected by peptide bonds.
4. Polysaccharides:
5. Monomer: The left section shows a saccharide with a basic carbon ring structure.
6. Polymer: The right section presents a polysaccharide chain with multiple saccharides linked together.

7. Polynucleotides:

8. Monomer: The left section features a nucleotide with a phosphate group, sugar, and base, highlighted in different colors for each part.
9. Polymer: The right section illustrates a polynucleotide chain with repeated sequences of nucleotides forming a DNA structure equivalent.

Each section is clearly labeled, indicating the monomer and polymer of each macromolecule type.

[Generated by AI]

Polypeptides

Polypeptides are polymers that make up proteins from monomers called amino acids. The general structure for an amino acid monomer is shown in **Figure 6**. Looking at the general structure of amino acids, what do you notice? Does the structure contain functional groups that we have already seen so far? Notice how the amino acid has two different functional groups. This means that we do not need two different types of monomers to create a polymer as we saw earlier, we can use just one monomer to create a polymer. Looking at the structure of the amino acid monomer, would you expect amino acids to form polyesters or polyamides? Let's look at the formation of a polypeptide in **Figure 7**.

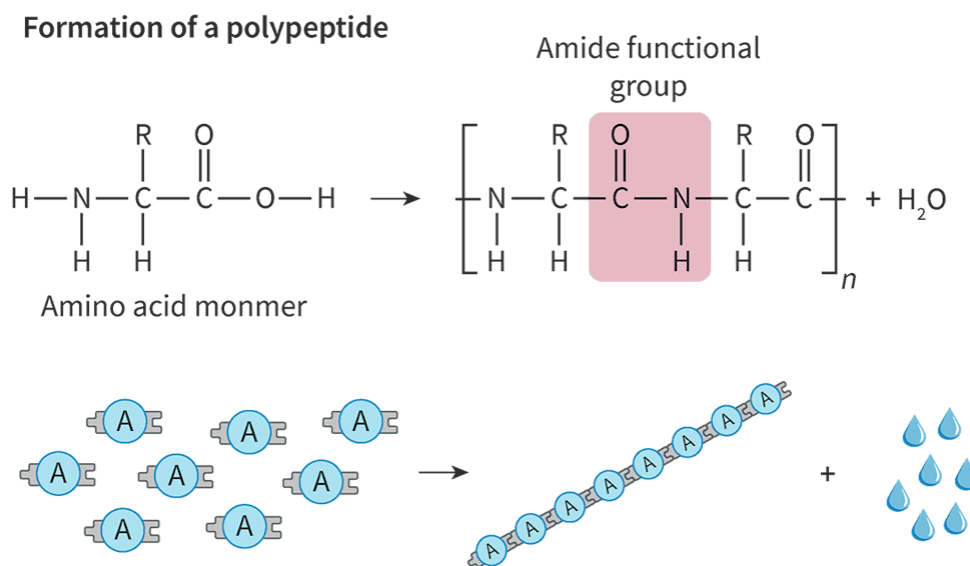


Figure 7. Condensation reaction to form a polypeptide.

More information for figure 7

The image is a diagram depicting a condensation reaction forming a polypeptide. On the left side of the diagram, individual amino acid molecules are illustrated, each represented by the letter 'A' inside blue circles, and attached to a rectangular shape representing the amino acid structure. The amino acids are shown aligned horizontally, indicating the start of a forming chain.

Arrows pointing from each amino acid toward a central bond formation area visually indicate the process of linking these amino acids together. In the center, a pink area highlights the formation of an amide bond (depicted by the chemical structure $\text{O}=\text{C}-\text{N}-\text{H}$), signifying the creation of a polypeptide bond between the amino acids.

To the right, the newly formed polypeptide chain is shown, represented by multiple 'A' circles connected in a linear sequence to emphasize the continuous structure of the polymer. Above the polypeptide, water molecules are illustrated as byproducts of the reaction, shown as small blue droplet icons.

This whole process demonstrates how a polypeptide forms as a polymer from amino acids via condensation reaction, with water being released as a byproduct, illustrated by the droplets and chemical bond formation.

[Generated by AI]

You can see in **Figure 7** that when amino acids combine to form a polypeptide polymer, the amide functional group is formed, making the polypeptide polymer an example of a polyamide. Notice how water is also produced as a byproduct during this condensation reaction. Proteins can contain polypeptides formed from anywhere from 11 to 35 000 amino acids.

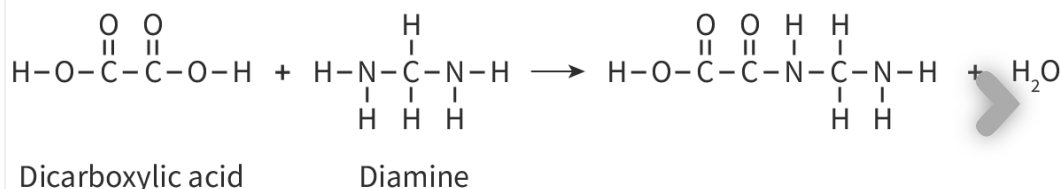
Polysaccharides

Polysaccharides are the polymers that make up sugars, such as starch and cellulose. Monomers called saccharides combine through a condensation reaction to form a polysaccharide and water. The structures of the polysaccharide starch and its saccharide glucose monomer are shown in **Figure 6**.

Polynucleotides

Polynucleotides are genetic material, such as your DNA, that form by a condensation reaction of monomers called nucleotides. The structures of a polynucleotide and its nucleotide monomer are shown in **Figure 6**. Nucleotides are made up of a base, a sugar and a phosphate component. The reacting functional groups are on the sugar and phosphate components. The genetic code is contained in the sequences of bases. DNA consists of two strands of polynucleotides twisted together and attracted by hydrogen bonds with the bases on the inside to protect the code. This is illustrated in **Figure 8**.

All biological macromolecules can be broken down by a process called hydrolysis. Hydrolysis is the process of breaking a molecule into two parts using a water molecule. The slides in **Interactive 2** show the processes for condensation and hydrolysis for polypeptides, polysaccharides and polynucleotides.



Interactive 2. Processes for condensation and hydrolysis for polypeptides, polysaccharides and polynucleotides.

 More information for interactive 2

The slideshow has six slides for the Processes for Condensation and Hydrolysis of Various Polymers.

Slide 1: A chemical equation showing the reaction between a dicarboxylic acid and a diamine. The dicarboxylic acid is represented as a molecule with two carboxylic acid groups ($-\text{COOH}$) on either end. The structure starts with a hydrogen atom bonded to an oxygen (hydroxyl group), followed by a carbon double bonded to an oxygen (carbonyl group). This carbon is connected to another carbon with the same carboxylic acid structure, and its structural formula is $\text{HOOC}-\text{COOH}$. The acid is labelled as Dicarboxylic acid under it. The diamine is depicted as a central carbon atom bonded to two amine groups ($-\text{NH}_2$) and two hydrogen atoms. The chemical formula of the amine is $\text{H}_2\text{N}-\text{CH}_2-\text{NH}_2$. In this case, the central carbon is shown with one amine on each side. The molecule is labelled as diamine under it. The product is given as $\text{NH}_2-\text{CO}-\text{COO}-\text{CH}_2-\text{NH}_2$ along with a water molecule. Each time an amine group reacts with a carboxylic acid group, an amide bond ($-\text{CONH}-$) is formed, and a molecule of water (H_2O) is released as a by-product.

Slide 2: The $-\text{OH}$ group on dicarboxylic acid and H on amine on the reactant side and water molecule on the product side are highlighted. The reaction is the same as the previous slide, and the formation of water molecule is highlighted. On-screen text reads: Water is formed from the $-\text{OH}$ of the functional group on the carboxylic acid combining with the H of the functional group on the amine.

Slide 3: A chemical equation showing polymer formation from dicarboxylic acid and diamine with reactive functional groups is highlighted in the product. The structural formula of the product is

$\text{H}_2\text{N} - \text{CO} - \text{CO} - \text{NH} - \text{CH}_2 - \text{NH}_2$. The carboxylic acid group and the amine group on the product are highlighted in pink to indicate the presence of remaining reactive sites that enable further polymerisation. On-screen text reads: Notice how the polymer formed still has the functional group of the amine and carboxylic acid. This allows it to react further.

Slide 4: The reaction shown involves the formation of a polyamide through a process called condensation polymerisation. In this reaction, monomers containing amine ($-\text{NH}_2$) groups and carboxylic acid ($-\text{COOH}$) groups react together with an amide. As shown in the first diagram, when the three molecules combine, two water molecules are produced, highlighted by the removal of H and $-\text{OH}$ groups. The resulting product is a chain of repeating units joined by amide bonds, with functional amine and carboxylic acid groups still present at the ends. This allows the molecule to continue reacting with additional monomers to form a longer polymer chain. On-screen text reads: In total now 3 H_2O molecules have been produced.

Slide 5: It highlights these reactive end groups, amine and carboxylic acid, demonstrating the potential for further polymer growth. On-screen text reads: The product formed still has the functional groups of an amine and carboxylic acid. This allows it to continue to react to form a polymer.

Slide 6: A dicarboxylic acid monomer reacts with a diamine monomer to form a polyamide polymer and water. The general reaction can be summarised as n dicarboxylic acid molecules reacting with n diamine molecules to form a polyamide and $(2n \text{ minus } 1)$ water molecules.

This video shows more detail about hydrolysis of polypeptides.

Amino Acids 12. Protein Hydrolysis.



Video 4. Hydrolysis of polypeptides.

 More information for video 4

1

00:00:00,880 --> 00:00:03,280

narrator:

Here I take a look at protein hydrolysis.

2

00:00:03,800 --> 00:00:05,480

By definition, hydrolysis means

3

00:00:05,560 --> 00:00:08,280

to split a molecule apart

using the elements of water.

4

00:00:09,080 --> 00:00:13,040

Typically, an alkali or an aqueous acid
is used to bring this about.

5

00:00:14,080 --> 00:00:16,280

Now as hydrolysis reactions go,

6

00:00:16,360 --> 00:00:19,240

the hydrolysis

of proteins are relatively difficult.

7

00:00:19,880 --> 00:00:22,560

Remember, proteins have at least a primary

8

00:00:22,720 --> 00:00:28,120

and secondary structure and often tertiary
and indeed quaternary structure.

9

00:00:28,920 --> 00:00:32,240

The first of two methods

to bring about protein hydrolysis

10

00:00:32,320 --> 00:00:35,720

would be to reflux the protein
with six molar hydrochloric acid

11

00:00:35,800 --> 00:00:37,000

for about 24 hours.

12

00:00:37,840 --> 00:00:42,680

And the second method

would be to use protease enzymes

13

00:00:43,160 --> 00:00:44,840

over a period of several hours.

14

00:00:45,000 --> 00:00:47,880

At a line body temperature,
37 degrees centigrade.

15

00:00:48,480 --> 00:00:51,200

And of course, the reason
that it takes such a long time

16

00:00:51,280 --> 00:00:54,440

and a lot of effort to bring
about the hydrolysis of a protein

17

00:00:55,120 --> 00:00:57,080

is because of its complex structure.

18

00:00:57,280 --> 00:00:59,880

You're going to have to break
down any quaternary structure,

19

00:01:00,320 --> 00:01:04,960

any tertiary structure,
any secondary structure

20

00:01:05,360 --> 00:01:08,240

before you get to this point
where you've got your polypeptide chain.

21

00:01:08,320 --> 00:01:11,040

So you're going to have
to break all those bonds

22

00:01:11,120 --> 00:01:15,200

that cause the twisting and folding,
namely Van der Waals' forces,

23

00:01:15,560 --> 00:01:20,000

electrostatic attractions,
hydrogen bonds, disulfide bonds

24

00:01:20,080 --> 00:01:21,640

before you get to this level.

25

00:01:22,320 --> 00:01:23,520

But what I'm going to show then

26

00:01:23,600 --> 00:01:25,320

is what happens
once you've got to this level.

27

00:01:25,400 --> 00:01:29,240

This is the hydrolysis

of the peptide links

28

00:01:29,720 --> 00:01:32,200

I'm showing there

with the scissors that the bond

29

00:01:32,280 --> 00:01:34,640

between the CO and the NH bricks.

30

00:01:34,920 --> 00:01:38,080

We put an OH to the left

giving us back the carboxyl group,

31

00:01:38,560 --> 00:01:41,400

and we put a H to the right,

giving us back our NH₂.

32

00:01:41,800 --> 00:01:44,960

So there's my first amino acid

molecule produced

33

00:01:45,080 --> 00:01:47,480

and this process

is repeated over and over.

34

00:01:47,720 --> 00:01:50,120

So let's bring in the scissors

35

00:01:50,360 --> 00:01:54,160

and bring about the hydrolysis

of the next peptide link.

36

00:01:54,680 --> 00:01:56,400

So again, this bond between

37

00:01:56,760 --> 00:01:59,120

the C double bond O

and the NH is going to break.

38

00:01:59,760 --> 00:02:02,520

We'll put the OH together

with the C double bond O

39

00:02:02,600 --> 00:02:04,200

to give back the carboxyl group.

40

00:02:04,720 --> 00:02:06,640

And we've got another amino acid there.

41

00:02:06,720 --> 00:02:10,840

And we'll put a H along with the NH

to give us back our NH₂.

42

00:02:11,760 --> 00:02:13,760

Now we've got two molecules of amino acid.

43

00:02:14,440 --> 00:02:17,840

Let's bring the scissors

in and snip another peptide link.

44

00:02:18,480 --> 00:02:22,080

The peptide link is made up

of the CO and the NH by the way.

45

00:02:22,960 --> 00:02:25,720

But in hydrolysis,

I'm showing this the bond

46

00:02:25,800 --> 00:02:27,320

between the C and the N break.

47

00:02:27,960 --> 00:02:31,160

And we've got another,

a third molecule of amino acids.

48

00:02:32,320 --> 00:02:36,640

So we've already broken

the quaternary structure,

49

00:02:36,720 --> 00:02:39,960

if it exists, we've broken

the tertiary structure if it exists.

50

00:02:40,040 --> 00:02:42,560

Remember, fibrous proteins

don't really have a tertiary

51

00:02:42,640 --> 00:02:44,240

or indeed quaternary structure.

52

00:02:44,600 --> 00:02:47,560

Globular proteins

do have a tertiary structure

53

00:02:47,760 --> 00:02:50,080

and often a quaternary structure as well.

54

00:02:51,560 --> 00:02:56,320

And what's holding these twisted

and folded structures together

55

00:02:56,680 --> 00:02:58,760

are generally four kinds of bonds,

56

00:02:59,680 --> 00:03:02,840

namely Van der Waals'

forces between, for example,

57

00:03:03,040 --> 00:03:06,120

alkyl R groups, electrostatic attractions,

58

00:03:06,200 --> 00:03:09,040

for example,

between lysine and aspartic acid,

59

00:03:09,480 --> 00:03:13,160

hydrogen bonds between,

for example, serine and aspartic acid

60

00:03:13,560 --> 00:03:16,880

and disulfide bonds

between cystine molecules.

61

00:03:17,920 --> 00:03:20,880

And we've nearly finished snipping these

62

00:03:21,240 --> 00:03:23,440

peptide links bringing about hydrolysis.

63

00:03:23,520 --> 00:03:24,800

There's our last one,

64

00:03:25,760 --> 00:03:28,840

and again,

we put an OH on the C double bond O.

65

00:03:28,920 --> 00:03:30,680

We put a H on the NH,

66

00:03:31,720 --> 00:03:36,440

and we brought about hydrolysis

creating 1, 2, 3, 4, 5, 6,

67

00:03:36,800 --> 00:03:39,320

now amino acid molecules.

68

00:03:40,280 --> 00:03:43,000

Finally, let's see how we could summarize

69

00:03:43,080 --> 00:03:45,880

this hydrolysis of the polypeptide chain.

70

00:03:46,840 --> 00:03:48,400

So if we take a protein

71

00:03:48,680 --> 00:03:52,960

and be represented

as n amino acids joined together,

72

00:03:53,200 --> 00:03:55,920

let's count how many

water molecules we'd need.

73

00:03:56,360 --> 00:03:59,040

Well, if you had

three amino acids joined together,

74

00:03:59,760 --> 00:04:03,320

you would require two water molecules

because you would've two peptide links.

75

00:04:03,520 --> 00:04:06,120

So the number of water molecules

is one less

76

00:04:06,200 --> 00:04:08,560

than the number

of amino acids in the chain.

77

00:04:09,320 --> 00:04:15,040

But what you can do is instead of having

awkwardly used $n - 1$ water molecules,

78

00:04:15,280 --> 00:04:17,640

you can have n water molecules used,

79

00:04:18,000 --> 00:04:20,040

if you take off

one additional water molecule.

80

00:04:20,120 --> 00:04:22,280

And I've done that by leaving the H off

81

00:04:22,360 --> 00:04:24,360

outside the bracket to the left

82

00:04:24,440 --> 00:04:27,160

and the OH off outside

the bracket to the right.

83

00:04:27,600 --> 00:04:30,040

So that's been my look

at protein hydrolysis.

84

00:04:30,320 --> 00:04:32,800

Do please consider looking
at others in the series,
85

00:04:33,000 --> 00:04:34,280

amino acids and proteins.

In this next activity you will use a folded piece of paper to model the formation of condensation polymers from their monomers. You will also design your own paper model for a condensation polymer.

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Combining different ideas in order to create new understandings
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pairs activity

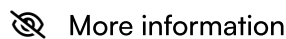
Print out the following page . If you do not have a printer request that your teacher print it out for you.

Page 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry%20IB%20DP%20Chemistry%20SL%20HL%20FE2025%20Condensation%20Polymers%20HL%20ID%2043847%20Print%20out%20the%20following%20page%20.pdf)

Cut out each polymer into its own strip. Each partner gets their own strip. Perform the following:

- Identify all of the functional groups in the monomer. Share this identification with your partner.
- Fold along the dotted lines to 'condense' the monomers into a polymer.
- Classify the polymer as either a polyester or polyamide. Share your classification with your partner.
- Deduce how many water molecules were condensed from the reaction of this number of monomers.
- Individually, make your own strip out of one of the polymers discussed in this section. You should select a different polymer than your partner.
- Together make a strip to represent the polypeptide shown below.



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